

## ***Diet throughout childhood and age at menarche in a contemporary cohort of British girls.***

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OBJECTIVE: To investigate associations between dietary intakes throughout childhood and age at menarche, a possible indicator of future risk of disease, in a contemporary cohort of British girls.

DESIGN: Diet was assessed by FFQ at 3 and 7 years of age, and by a 3 d unweighed food diary at 10 years. Age at menarche was categorised as before or after 12 years 8 months, a point close to the median age in this cohort. SETTING: Bristol, South-West England. SUBJECTS: Girls (n 3298) participating in the Avon Longitudinal Study of Parents and Children.

RESULTS: Higher energy intakes at 10 years were positively associated with the early occurrence of menarche, but this association was removed on adjusting for body size. Total and animal protein intakes at 3 and 7 years were positively associated with age at menarche  $\leq 12$  years 8 months (adjusted OR for a 1 sd increase in protein at 7 years: 1.14 (95 % CI 1.04, 1.26)). Higher PUFA intakes at 3 and 7 years were also positively associated with early occurrence of menarche. Meat intake at 3 and 7 years was strongly positively associated with reaching menarche by 12 years 8 months (OR for menarche in the highest v. lowest category of meat consumption at 7 years: 1.75 (95 % CI 1.25, 2.44)).

CONCLUSIONS: These data suggest that higher intakes of protein and meat in early to mid-childhood may lead to earlier menarche. This may have implications for the lifetime risk of breast cancer and osteoporosis.